Shaurya Tiwari

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SKILLS

Programming Languages: C/C++ (CUDA), C#, .NET, Java, Python, Rust, JavaScript, TypeScript, SQL, Swift, PHP, R
Data Science Tools: Python Libraries (OpenCV, PyTorch, TensorFlow, seaborn, Numpy, Pandas, Pillow)
Web/Database: JavaScript (Node.js, React), SQL Databases (MySQL, PostgreSQL), Django REST framework for RESTful APIs
Cloud Computing: Microsoft Azure (Virtual Machines (VMs), Load Balancer, Traffic Manager), GCP (Compute Engine, Cloud Storage), AWS(EC2, S3)
Additional Skills: DevOps (Docker, Kubernetes, Jenkins, GitLab CI/CD), Version Control (Git, Bitbucket), MLOps, NLP, LangChain, RAG

EXPERIENCE

Center for Ocean-Atmospheric Prediction Studies:

Software Engineer | (Tallahassee, Florida) May 2023 - May 2024

- Developed **Data Pipeline** using **Python** in **Apache Airflow** for marine data extraction, processing of over **1 million** data points.
- Implemented a scalable **PostgreSQL** database architecture, for the storage and management of **10+ TB** of oceanographic data.
- Designed a backend **API** using **Django REST Framework (DRF)**, facilitating data access for **2M+** users, increasing data utilization.
- Built interactive web interface with **React.js** and Plotly.js, enabling users to access and analyze complex **oceanographic** data.

Department of Scientific Computing, FSU:

Graduate Research Assistant | (Tallahassee, Florida) August'2022 - April'2023

- Developed **Generative Adversarial Networks (GANs)** utilizing libraries **PyTorch and TensorFlow** for deep learning models from inception.
- Achieved **0.92 F1-score** and **0.95 AUC-ROC** with deep learning model for real-time environmental anomaly classification.
- Model robust to noise, handled multimodal inputs, and detected deviations with **0.90 precision** and **0.94 recall**.
- Optimized deep learning pipeline using custom **CUDA GPU** kernels in **C++**, reducing processing time by **40%** for anomaly detection.
- Enhanced computational speed by **30%** through GPU clustering and applied **HPC** techniques for scalable neural network training.

ACS Pvt. Ltd:

DevOps Engineer | (Mumbai, India) January'2021 - December'2021

- Designed **CI/CD pipeline** using Jenkins, SonarQube, Docker; resolved 500+ code issues and security vulnerabilities from inception.
- Deployed CI/CD pipeline and containerized apps on **AWS**, utilizing **EC2** for compute and **S3** for storage.
- Integrated **AWS CloudWatch** for monitoring; set alarms to cut mean time to resolution by **80%** through automated actions.

Cart Geek:

Web Development - Intern | (Mumbai, India) June'2020 - December'2020

- Deployed and managed **Microsoft Azure** based hosting with **.NET** applications using App Services and Traffic Manager for optimization.
- Leveraged **ASP.NET Core** for building and deploying scalable web applications and APIs, ensuring high availability and performance.
- Utilized **Google Cloud Platform**: **Compute Engine** and **Cloud Storage**, to host dynamic web applications and manage data sets securely.

SELECTED PROJECTS

Intelligent AI Chatbot for Personal Portfolio | AI Application Project May' 2024 - Ongoing

- Developed an intelligent chatbot for my portfolio site, enabling users/recruiters to discuss my professional career in natural language, trained on custom data like resume and accolades. Utilized **Llama 3** (open-source language model), **Retrieval-Augmented Generation** (to fetch customized queries), **Chroma Vector DB** (storage and retrieval of embeddings), and **LangChain** (open-source framework).

Learning/Teaching concepts related to LLM | YouTube Portfolio, (Channel Link) March 2024 - Ongoing

- Through my YouTube channel, I've facilitated learning in Data Science, Machine Learning, and Deep Learning, using **Python** for scripting to clear concepts surrounding **Large Language Models** by building transformer models on customized textual data.

Built a Text-Completion model - Transformer Architecture | CIS 5930, (FSU) Feb 2024 - Apr 2024

- Engineered a **text completion model [Python]** from ground up based on the **transformer** (self-attention) architecture as shown in the famous research paper: Attention is all you need, concepts of which are the core foundation of **LLMs** like GPT-3, BERT, and BART.

Polyp Segmentation using Deep Learning | COP 5725, (FSU) Aug' 2023 - Dec' 2023

- Developed robust deep learning polyp segmentation model using **Python** and its deep learning frameworks like **PyTorch, OpenCV,** and various libraries. Achieved accurate polyp segmentation from colonoscopy images, achieving high Dice Similarity Coefficient (DSC) of 0.92 and Jaccard Coefficient (JC) of 0.86 while being robust to variations in **polyp size, shape, and location**.

EDUCATION

Florida State University, *MSc in Computer Science* | Florida, USA GPA: 3.8 / 4.0 May 2024
NMIMS University, *B.Tech in Information Technology* | Mumbai, India GPA: 3.3 / 4.0 Aug 2022

RELEVANT COURSES: Data Mining | Machine Learning | Data Structure Algorithm | Database Systems | Artificial intelligence